THE UNIVERSITY of EDINBURGH

Centre for Cognitive Ageing and Cognitive Epidemiology

The Effect of Age on Visual Working Memory for Features and Conjunctions

Stephen Rhodes[1], Mario A. Parra[1], Nelson Cowan[1,2], & Robert H. Logie[1]

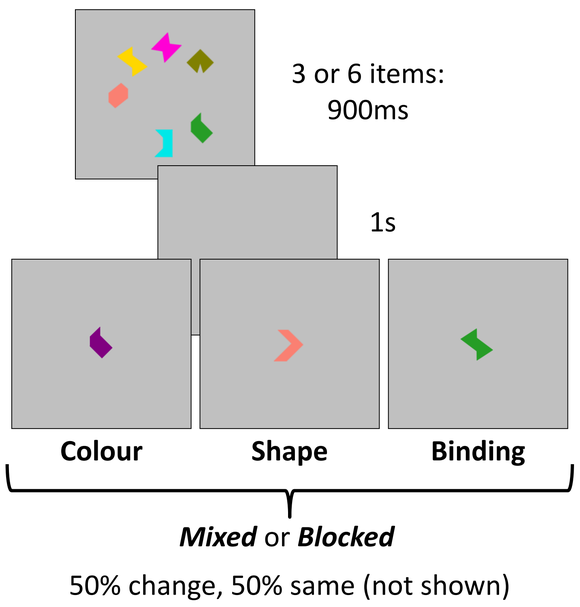
[1]University of Edinburgh, [2]University of Missouri-Columbia

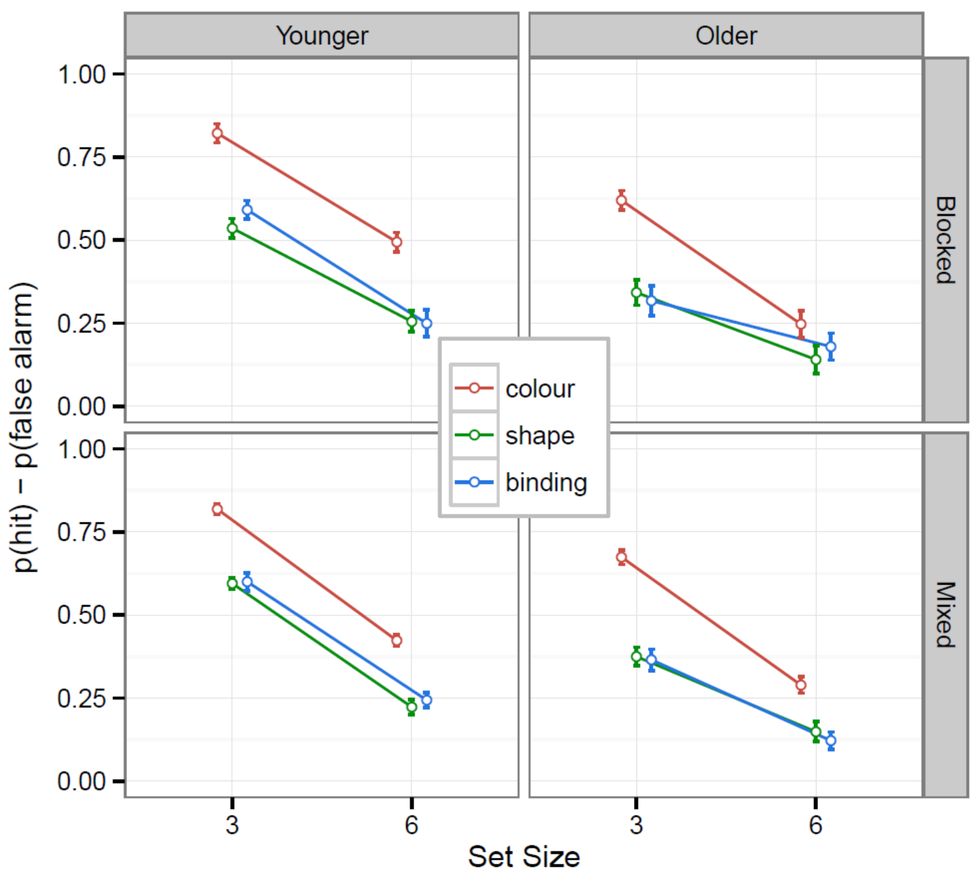
Introduction

- Visual working memory (VWM) displays substantial decline across the adult life-span [1].
- *Older adults exhibit a clear associative deficit [2, 3] – some VWM decline may be due to impaired ability to bind object features into a unitised representation [4].*
- *However, VWM for colour-shape conjunctions appears no more age-sensitive than individual features [4-6].*
- Conjunctive binding deficit appears to differentiate early Alzheimer's disease from healthy ageing [7, 8].
- There remain boundary conditions where older adults may struggle to detect conjunction changes [9, 10].
- *Cowan et al. [10]: small colour-location binding deficit when trials were blocked. Mixing colour and binding changes resulted in large binding deficit – older adults missed binding changes.*
- *As previous studies have only used blocked trials Experiment 1 assessed whether this is true for colour-shape conjunctions.*
- To quantify evidence *against* age-group by memory condition interactions we report default Bayes factors [11]†.

Experiment 1

- Does mixing feature and conjunction changes affect older adults' ability to detect changes to colour-shape binding?
- **Method:** Change detection task. 48 younger (20.92 ± 2.76) and 48 heathy older adults‡ (70.61 ± 5.00). Half saw blocked, half mixed.





Clear effect of age (BF > 100). But evidence against age by memory condition interactions∓.

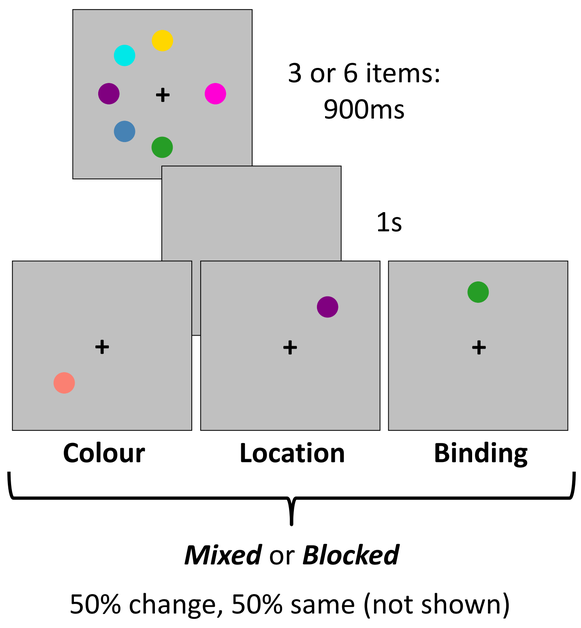
- Age-group × memory condition: *BF* = 16.7
- Age × condition × block type: *BF* = 5.0
- 4 way interaction: *BF* = 4.0

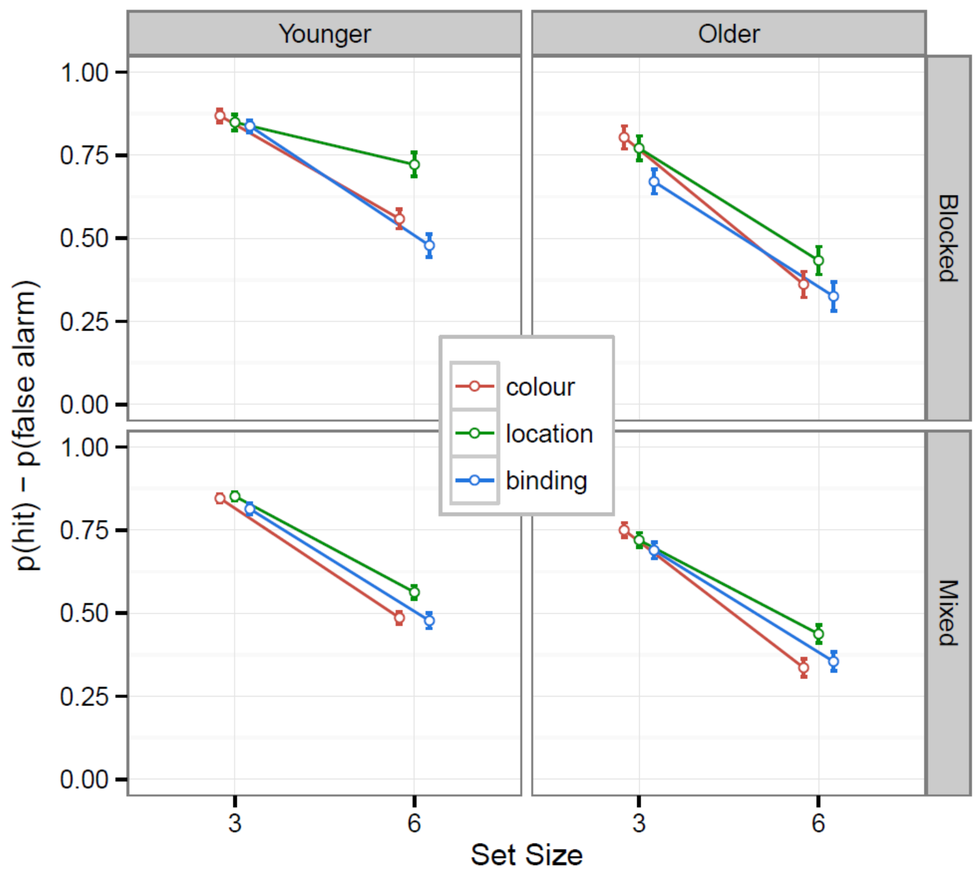
Slight evidence for age × condition × set size: BF = 3.1

- For older adults effect of set size smaller for shape & binding

Experiment 2

- Is binding to location a specific problem?
- Cowan et al. [10] included colour and binding changes only.
- Is there a colour-location binding deficit when equal weight given to both individual features?
- **Method:** 48 younger (20.92 ± 2.15) and 48 older adults (70.81 ± 4.70). Half saw blocked, half mixed.





Again, clear effect of age (BF > 100). But evidence against age by memory condition interactions.

- Age-group × memory condition: *BF* = 17.5
- Age × condition × block type: *BF* = 11.6
- Age × condition × set size: *BF* = 2.1
- 4 way interaction: *BF* = 1.8. (colour vs binding: *BF* = 2.8)

Discussion

- *We find substantial-strong evidence against a differential effect of age on VWM for features and conjunctions.*
- *Mixing feature and binding trials did not affect older adults' ability to distinguish conjunction changes.*
- Experiment 1 adds to growing body of evidence against colour-shape binding deficit [4-6].
- Experiment 2 adds to mixed findings regarding object-location binding deficits. Some find older adults less able to detect object-location changes [12, 13], whereas others have not [14].
- Future work should address factors contributing to presence/ absence of object-location deficit. Possibly stimulus complexity.
- The absence of age-related binding deficit for within-object (intrinsic) features is in stark contrast to the deficit found for between-object (extrinsic) associations in LTM [2] and WM [3].
- The magnitude of age-related binding deficit may depend on extent to which feature conjunctions are represented as a unitised object [e.g. 15]. Future work should directly compare the effect of age on intrinsic and extrinsic binding in VWM.

Notes: † Bayes factors were calculated using the `BayesFactor` package (http://CRAN.R-project.org/package=BayesFactor) in `R` (http://www.R-project.org/). ‡ All older adults scored over 27 on the MMSE. ∓ For both experiments our conclusions are not changed by analysing *d'*, *A'*, hit rate, or proportion correct instead of hits minus false alarms (*Pr*). Further, there was no differential effect of age on measures of bias (*Br*, *c*, *B''*) for features and conjunctions.

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Acknowledgements: This work was undertaken by the University of Edinburgh MRC Centre for Cognitive Ageing and Cognitive Epidemiology, part of the cross council Lifelong Health and Wellbeing initiative. Funding from the BBSRC, EPSRC, ESRC and MRC is gratefully acknowledged (G0700704).

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